



Okay so my research sub agents have gone through my 93 downloaded papers and they came up with the following stuff so what should I do now and And verify their correctness, and validate the whole thing, and give me my next steps:

Dynamic workflow "Multi-agent deep literature synthesis across 93 papers to formulate optimal thesis scope" completed · 15m 15s

Read 1 file, ran 4 shell commands

Executive Verdict on Initial Abstract

The initial abstract attempted to combine five distinct PhD-scale problems into a single undergraduate proposal. The 93-paper corpus analysis revealed three fatal flaws alongside three high-impact opportunities.

Fatal Flaws in the Initial Proposal

- Heartbeat Biometric Identification is Physically Impossible on Commodity WiFi. Myocardial apex displacement is approximately 0.2 to 0.5 mm. Chest breathing displacement is 5 to 12 mm. This creates a 30 to 40 dB signal-to-noise ratio deficit where cardiac micro-Doppler is completely masked by respiratory harmonics and ambient multipath. No peer-reviewed paper demonstrates unconstrained occupant identification from commodity single-link heartbeat signals. Reviewers at ACM MobiCom or IEEE INFOCOM will reject this claim.
- Closed-Set Classification Disguised as Intruder Detection. Training a Softmax classifier on five known volunteers does not test security. A real intruder is an unseen individual from outside the training distribution. The system must perform open-set rejection.
- The Base-Rate Fallacy in Elderly Inactivity Monitoring. In a residential setting, an elderly resident spends hours sitting quietly, reading, or napping. Simple threshold alerts on movement absence produce dozens of false alarms every day. Respiration liveness must serve as the physical gating signal to separate safe rest from medical distress.

High-Impact Defensible Opportunities

- Auditing and Solving the Benchmark Data Leakage Crisis. Recent audits (Sensors 2024, Varga) prove that widely cited CSI benchmarks leak data when sliding windows are randomly shuffled before splitting. Enforcing strict Leave-One-Subject-Out (LOSO) and

Leave-One-Environment-Out (LOEO) protocols provides an immediate publishable contribution.

- Single-Link Dual-Antenna CSI Ratio Invariance. Multi-receiver setups like Widar 3.0 require six synchronized WiFi nodes. A consumer setup uses a single router. Dual-antenna CSI ratioing cancels transceiver phase drift (CFO and SFO) on a single commodity link.
- Hierarchical Finite-State Machine with Uncertainty Abstention. When WiFi signals are corrupted by high noise or walking, the vital sign head abstains. When motion stops, the system transitions to respiration tracking.

Recommended Thesis Title and Abstract

Title:

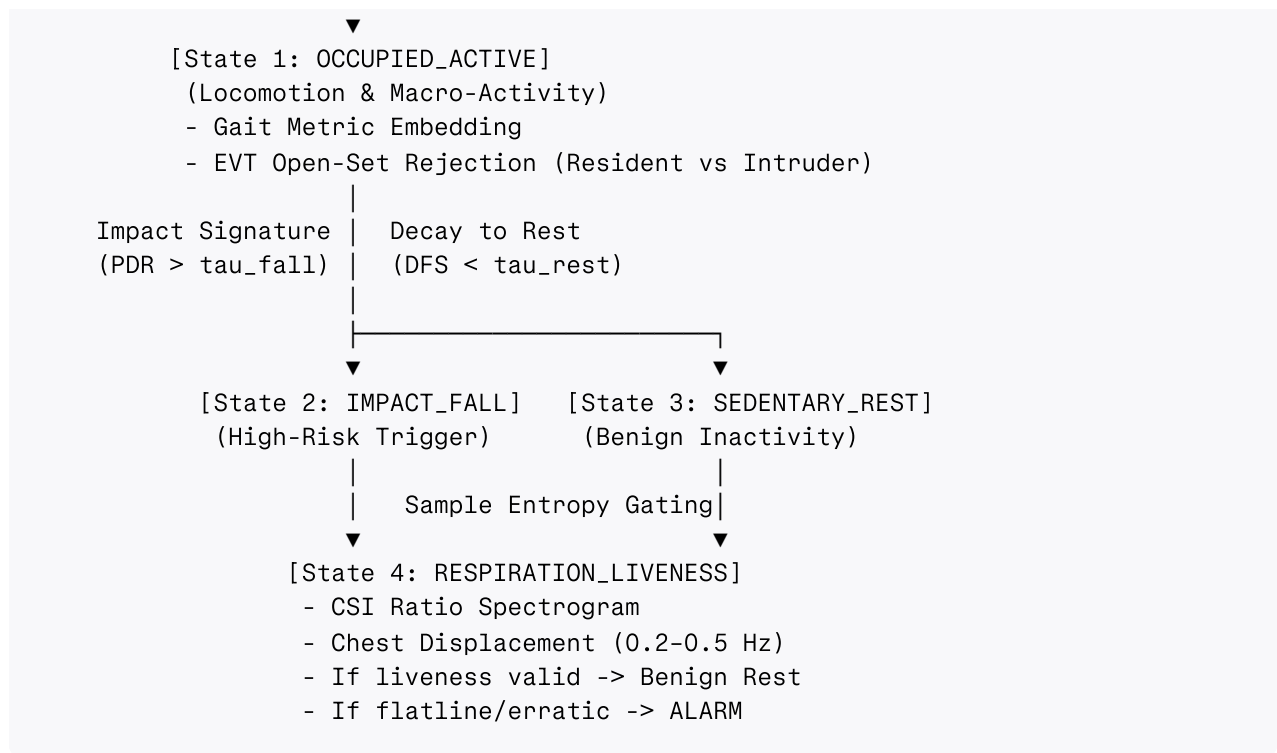
Physics-Guided, Zero-Leakage WiFi CSI Sensing for Residential Home Safety: Cross-Domain Activity Recognition, Uncertainty-Gated Inactivity Monitoring, and Open-Set Gait Verification on Constrained Edge Devices

Abstract:

Device-free ambient sensing via commodity WiFi Channel State Information (CSI) achieves high in-laboratory accuracy, but real-world residential deployment suffers from performance collapse across unseen environments and subjects, severe data leakage in academic benchmarks, and high false-alarm rates in inactivity monitoring. Furthermore, widely cited claims regarding passive cardiorespiratory biometric identification are invalid on commodity single-link hardware due to a 30 to 40 dB signal-to-noise ratio deficit and respiratory harmonic masking. This thesis presents a physics-grounded, zero-leakage ambient sensing framework tailored for resource-constrained edge execution. First, transceiver phase jitter and static multipath dependence are eliminated on single-link setups using dual-antenna CSI ratioing and conjugate Doppler cross-multiplication, paired with Subject-Adversarial Invariance Regularization to suppress biometric shortcut learning. Second, the base-rate false-alarm crisis in elder-care inactivity monitoring is resolved through a 5-state hierarchical Finite-State Machine: the framework establishes a calibration-free empty-room baseline via statistical autocorrelation testing, validates fall-impact transitions using the Power Decline Ratio, and disambiguates benign sedentary rest from emergency immobility via sample-entropy-guided respiration liveness tracking. Third, unviable vital-sign biometrics are replaced with few-shot open-set resident verification using dynamic locomotion gait kinematics trained via deep metric learning and gated by Extreme Value Theory distance ratios. Evaluated under strict Leave-One-Subject-Out and Leave-One-Environment-Out protocols across public benchmarks (CSI-Bench, Widar 3.0, WiMANS, SenseFi), compact Temporal Convolutional Networks and selective state-space models (BabyMamba, sub-50k parameters) maintain real-time edge execution within the memory constraints of an ESP32 microcontroller and a Raspberry Pi 4.

The 5-State System Architecture





The synthesis gives you a useful direction, but it is not yet a validated research specification. Keep the emphasis on cross-domain evaluation, open-set recognition, and uncertainty; correct the physics claims, redesign the safety logic, and establish hardware/dataset feasibility before committing to the architecture.

I audited your supplied report and the attached program specification against primary papers and documentation. This is a verification of their major claims—not a certification that all 93 PDFs were correctly synthesized. The attached specification currently turns several unproven assumptions into exclusion rules and planned conclusions. ^[1]

I’ve also prepared an 18-claim audit CSV with verdicts, corrected wording, sources, and required checks.

1. Claim-by-claim verdict

Agent claim	Verdict	What the evidence actually supports
Commodity-WiFi heartbeat biometrics are “physically impossible”	Unsupported absolute	Constrained heartbeat-rate estimation has been demonstrated on a stand-alone commodity WiFi link. That does not establish reliable heartbeat-based identity, but it invalidates a blanket impossibility argument. ^[2]
Cardiac sensing has a universal 30–40 dB SNR deficit	Not established	The supplied displacement numbers do not determine received CSI SNR. A link-budget/noise model and measurements are missing.
Closed-set classification is not sufficient for intruder detection	Correct	You need rejection of unenrolled identities. CAUTION is relevant open-set gait-recognition prior art. ^[3]

Agent claim	Verdict	What the evidence actually supports
Respiration separates safe rest from medical distress	Incorrect and unsafe	A person can be unresponsive while breathing normally. Breathing evidence must not automatically cancel a suspected-fall or welfare alert. [4]
Varga 2024 establishes benchmark evaluation problems	Correct, but overgeneralized	The retrieved paper audits subject-overlap in CSI-HAR, containing 420 samples from three volunteers—not specifically shuffled overlapping windows across all named benchmarks. [5]
Strict LOSO/LOEO provides an immediate publishable contribution	Overstated	Correct evaluation is necessary. Novelty needs a new, consequential finding, broader reproducible audit, dataset, or method beyond an already published correction. [5]
Widar3.0 requires six synchronized nodes	Incorrect as written	The original paper specifies one transmitter and at least three receivers; its experimental deployment uses six receivers. [6]
Dual-antenna ratio cancels phase drift	Correct under assumptions	It can cancel shared packet-dependent phase errors when antenna measurements share the relevant clock/timing errors. Residual noise and antenna-specific offsets remain. [2] [7]
Ratioing eliminates static multipath dependence	Incorrect	Hardware-offset cancellation does not remove propagation geometry or static multipath from the channel ratio. [6] [7]
Quiet for 60 seconds implies vacancy	Invalid inference	A stationary occupant can remain present. “No detected motion” is not sufficient evidence of an empty room; respiration sensing itself has geometry-dependent blind spots. [7]
Power Decline Ratio verifies a fall impact	Overstated	It is a candidate CSI feature, not direct proof of bodily impact or injury. Fall detection still needs classification and confounder testing. [8] [9]
Sample entropy validates physiological liveness	Unverified	Require the exact supporting paper, formula, thresholds, and reference-sensor experiment. Do not promote it to a validated medical discriminator.
All four benchmarks validate the entire framework	Incorrect	They contain different representations, labels, acquisition protocols, and tasks. Several cannot test your proposed front end or continuous safety logic. [10] [11] [3]
BabyMamba demonstrates ESP32 feasibility	Partly correct	Its updated preprint reports ESP32 deployment, but for inertial-sensor HAR—not your complete CSI acquisition, preprocessing, respiration, and identity pipeline. [12]

The central problem is that the agents have combined legitimate findings with stronger claims that those findings do not establish.

Heartbeat: remove the claim, not the literature

PhaseBeat reports breathing and heart-rate estimation using commodity WiFi and receiver-antenna phase differences. It explicitly describes a stand-alone WiFi link and uses a directional transmitting antenna for heart-rate monitoring because the cardiac effect is extremely weak. Its experiments involved four people; the reported median heart-rate error was 1.19 beats/min under its experimental conditions. [\[2\]](#)

This supports three separate conclusions:

- Heartbeat-related sensing is not universally physically impossible.

- Constrained heart-rate estimation is not equivalent to recovering a stable, identity-specific cardiac waveform.
- Reliable, unconstrained heartbeat-based occupant identification remains unestablished by the evidence reviewed here.

Replace the agent’s exclusion rule with:

“Unconstrained heartbeat-based identity recognition is outside scope because its longitudinal, cross-environment reliability has not been established for our intended hardware and operating conditions.”

Retain those papers in the literature review. Otherwise, you risk excluding counterevidence to your own argument.

The proposed decibel reasoning also needs correction. Using the agents’ displacement ranges, a simplified amplitude contrast would be:

$$20 \log_{10} \left(\frac{d_{\text{breathing}}}{d_{\text{cardiac}}} \right) \approx 20\text{--}36 \text{ dB}.$$

That is an illustrative displacement-derived contrast under a proportional-response assumption—not a measured CSI SNR deficit. It does not account for reflection strength, geometry, receiver noise, respiratory harmonics, or processing gain.

Leakage: distinguish two problems

Varga’s December 2024 paper examines the CSI-HAR dataset from Moshiri et al. It trains on two participants and tests on the third, reporting performance reduced to approximately 63–72% of the originally reported accuracy. That is a specific subject-generalization audit, not evidence that every CSI benchmark suffers the same temporal-overlap problem. ^[5]

Your audit must distinguish:

- **Actual information leakage:** shared raw samples, overlapping windows across splits, duplicates, test-fitted preprocessing, or test-driven model selection.
- **Evaluation mismatch:** testing new recordings from known users while claiming performance on unseen users.

Known-user evaluation is legitimate when known users are the intended deployment population. It simply does not establish unseen-user generalization. For identity verification, enrolled users necessarily appear in enrollment and genuine-user testing; separation must occur across sessions, days, and relevant environments.

This corrects an oversimplification in my earlier guidance too: participant separation is not a universal requirement for every CSI task.

2. Architecture and hardware corrections

Fix the safety inference first

Your current architecture effectively contains two implications:

Respiration detected $\Rightarrow$ Benign rest

Respiration not detected $\Rightarrow$ Emergency

Neither implication is justified.

An unresponsive person can continue breathing, while a healthy breathing person can produce an unusable CSI signal because of position, orientation, interference, or a sensing blind spot. ^[4]
^[7]

For example, a person who falls, cannot get up, and continues breathing still warrants a welfare check. Conversely, a person sitting safely outside the sensitive geometry must not be classified as experiencing respiratory arrest.

Use the wording “respiration evidence” or “respiration observability,” not “proof of safety.”

Replace the five-state logic

The current FSM mixes occupancy states, events, sensing operations, and clinical interpretations. “Impact fall” is an event hypothesis; “respiration tracking” is an operation; “sedentary rest” is already an interpretation.

I recommend these five operational states:

State	Meaning	Permitted behavior
Unknown	Startup, ambiguous occupancy, sensor failure, or insufficient evidence	Abstain; report sensing uncertainty
Vacant	Available evidence supports an empty monitored area	Monitor for entry/activity
Occupied active	Occupancy supported and macro-motion detected	Activity sensing; optional gait verification
Occupied stationary	Occupancy supported with little macro-motion	Assess respiration observability; estimate rate if usable
Check required	Suspicious event or unresolved welfare condition	Request human verification; do not claim diagnosis

Track these as parallel variables rather than additional physical states:

- Signal quality.
- Suspected-fall event and its timestamp.
- Time since last movement.
- Respiration evidence and confidence.
- Whether occupancy is unresolved.

- Human acknowledgment.

Proposed decision rules:

1. A quiet timeout moves active occupancy toward stationary occupancy—not directly to vacancy.
2. Missing respiration with poor measurement quality produces “unobservable.”
3. Detected respiration does not clear a suspected-fall event.
4. A suspected fall followed by prolonged inactivity can trigger “check required,” even with detected breathing.
5. Vacancy requires supporting exit/occupancy evidence, not merely low Doppler energy.
6. Startup begins as unknown; you cannot assume the room starts empty.

The 60-second threshold should be an experimentally evaluated policy parameter, not a literature-established safety constant.

Ratioing is not domain invariance

Under an idealized shared-phase model,

$$\tilde{H}_1(k, t) = H_1(k, t)e^{j\phi(k, t)}, \quad \tilde{H}_2(k, t) = H_2(k, t)e^{j\phi(k, t)},$$

the ratio cancels the common phase:

$$\frac{\tilde{H}_1}{\tilde{H}_2} = \frac{H_1}{H_2}.$$

But if $H_i = S_i + D_i(t)$, with static and dynamic channel components,

$$R(k, t) = \frac{S_1(k) + D_1(k, t)}{S_2(k) + D_2(k, t)}.$$

The static channel terms remain. Therefore, ratioing can stabilize a measurement without making it independent of room, location, or orientation. FarSense proposes ratioing to improve respiration sensing and exploit amplitude/phase complementarity; Widar3.0 explicitly treats primitive Doppler features as domain-dependent.^{[6] [7]}

Also, ratio and conjugate multiplication are not interchangeable:

$$H_1 H_2^* = \frac{H_1}{H_2} |H_2|^2.$$

Compare them as alternative front ends. Do not stack them merely because both sound physics-guided.

Test:

- Raw amplitude.
- Inter-antenna phase difference.

- Regularized CSI ratio.
- Conjugate product.

Measure phase stability, denominator-related instability, respiration availability, cross-room accuracy, and preprocessing cost.

Resolve the ESP32 mismatch

A FarSense-style antenna ratio requires suitable coherent antenna measurements. Espressif’s documentation describes external antenna switching/selection; that is not equivalent to simultaneously recording two coherent receive chains. CSI-Bench also describes its ESP32-S3 configuration as 1×1.^{[10] [13]}

Two independent ESP32 receivers do not automatically share the common phase errors assumed by the ratio derivation.

Choose explicitly:

Option	Acquisition	Processing
Coherent-ratio research	Verified multi-RX-chain CSI receiver	Raspberry Pi/PC first; later embedded inference
ESP32-native research	Single-chain ESP32 CSI	Amplitude-based front end and compact model
Split deployment	Multi-antenna receiver produces features	ESP32 performs inference on those features

The third option is valid, but describe it as split acquisition/inference—not an ESP32-only sensing system.

Separate invariance from identity

Your specification simultaneously proposes suppressing subject information and using gait to identify subjects. That is a conflicting objective if both tasks depend on the same fully subject-invariant representation.^[1]

Use separate representations:

- Activity/safety branch: test subject-invariance regularization.
- Identity branch: preserve distinguishing gait information while testing room/day/layout robustness.

Start with separate encoders. Share components only if experiments demonstrate that sharing helps.

“Subject-Adversarial Invariance Regularization” should remain a working label until the agents provide its exact prior-art source or identify it explicitly as your proposed variant.

3. Dataset and model suitability

Do not download every dataset and assume it supports every claim. First build a capability matrix from the actual released files.

Resource	Appropriate use	Important limitation
Widar3.0 raw CSI	Cross-domain gesture evaluation; single-link front-end experiments using selected receiver data	Original BVP reconstruction is multiview; gesture clips do not validate continuous welfare monitoring. ^[6]
SenseFi packaged Widar	Model implementation and BVP-classification baselines	Its Widar input is processed BVP, not raw complex CSI; using it would not demonstrate a single-link front end. ^[3]
CSI-Bench	Classification, motion-source confounders, available OOD splits	Described representation is amplitude-only; it does not validate complex antenna-ratio preprocessing. Breathing is binary detection, not clinical safety assessment. ^[10]
WiMANS	Activity, occupancy/count-related extensions, multi-user stress tests	Three-second clips; nine everyday activities; no falls or physiological reference waveforms. ^[11]
NTU-Fi HumanID through SenseFi	Initial gait-identity experiments	Fourteen identities in the packaged benchmark; session/environment provenance must be verified before robust open-set claims. ^[3]
Your own continuous recordings	FSM, stationary presence, respiration observability, false alerts/hour	Requires synchronized references, consent, timestamps, and genuinely held-out sessions

Two discoveries materially change your plan:

CSI-Bench’s scale is task-specific. The whole collection covers 35 users and 26 environments, but its breathing subset lists only three users and three environments. Its paper already defines cross-user, cross-environment, and cross-device OOD settings for the multitask portion. “Adding cross-domain splits” alone is therefore not sufficient novelty. ^[10]

SenseFi is not an independent dataset. It is a library/benchmark covering underlying datasets, including Widar. Counting “Widar plus SenseFi” as two independent validations risks double-counting the same data. ^[3]

BabyMamba is real—but version-sensitive

The updated BabyMamba-HAR preprint, version 2 dated May 18, 2026, reports:

- Approximately 27K parameters for its crossover architecture.
- A handcrafted mixed-precision C++ runtime.
- Mean ESP32 inference latency of 154.4 ms across its eight benchmarks.
- Evaluation on inertial-sensor datasets, including UCI-HAR, MotionSense, PAMAP2, and Opportunity. ^[12]

Version 1 instead says microcontroller quantization remains uncharacterized. Your evidence ledger must record the exact version; otherwise agents can produce apparently contradictory findings from the same paper. ^[14] ^[12]

There is also an internal reporting discrepancy: version 2's abstract/text states an average F1 of 86.52%, while Table II lists 85.02% for the crossover model. Treat its numerical claims as requiring reproduction or author clarification, not unquestioned ground truth. ^[12]

Use a small causal TCN as the first deployable baseline. Add BabyMamba only if it improves an accuracy-latency-memory tradeoff on your CSI data.

Its bidirectional processing also means you must distinguish completed-window inference from immediate streaming decisions. Report window-acquisition time, preprocessing time, inference time, and alert-policy delay separately. ^[12]

4. Revised research specification

The agents have not actually reduced your scope enough. Your program still contains five substantial tracks: front-end physics, benchmark auditing/generalization, welfare decisions, identity, and embedded deployment. ^[11]

My recommendation is one primary contribution, one supporting deployment study, and one optional extension.

Recommended title

Quality-Aware Cross-Domain WiFi CSI Sensing for Residential Inactivity Monitoring

Add “on Edge Devices” only after end-to-end profiling establishes feasibility.

Avoid “zero-leakage” in the title. You can document leakage controls, but an absolute guarantee is harder to defend than “session-disjoint, subject-held-out, and environment-held-out evaluation.”

Primary research question

Can measurement-quality-aware selective prediction reduce false welfare-check alerts during stationary occupancy in unseen rooms without suppressing detection of suspicious inactivity events?

This combines your strongest ideas into a testable problem.

Core scope

- Single-person monitored area.
- Empty, active, and stationary occupancy.
- Respiration observability under stationary conditions.
- Selective prediction/abstention.
- Continuous event-level evaluation.
- Raspberry Pi deployment first.

Optional extension

- Open-set gait verification during usable walking segments.
- Separate identity representation and protocol.
- Output “unenrolled/unknown,” not “intruder.”

An unknown person could be a guest, caregiver, or technician. CSI identity rejection does not establish authorization or malicious intent; those require a separate policy.

Revised proposal abstract

Commodity WiFi channel state information offers a contactless modality for residential activity and respiration sensing, but deployment is challenged by environmental shift, weak physiological observability, and evaluation protocols that do not reflect continuous monitoring. This thesis will investigate a quality-aware framework for motion and stationary-occupancy monitoring across unseen environments. Where acquisition hardware permits coherent multi-antenna measurements, phase-difference, CSI-ratio, and conjugate-product representations will be compared with amplitude-based baselines. A selective sensing module will distinguish usable respiration evidence from ambiguous or degraded measurements, while a temporal decision policy will separate occupancy estimation, suspicious-event detection, and requests for human verification. Evaluation will use recording-disjoint partitions and held-out subjects and environments where supported by dataset metadata, supplemented by continuous recordings with synchronized reference measurements. Performance will be assessed using event recall, false alerts per monitoring hour, detection delay, respiration-estimation availability, and coverage-risk tradeoffs. End-to-end latency, memory, and power will be measured on selected edge hardware. Open-set gait verification may be investigated as a separate extension.

This abstract proposes research rather than presenting unperformed experiments as established results.

What would make it novel?

Treat these as candidate contributions, not confirmed gaps:

Candidate	Evidence you must produce
Quality-aware monitoring under shift	Improvement over fixed thresholds and confidence-only abstention in held-out rooms
Continuous decision policy	Lower false alerts/hour at matched suspicious-event recall
Physics-aware front end	Added benefit beyond amplitude features under the same folds and model
Reproducible benchmark audit	Newly documented failures, corrected artifacts, and conclusions beyond Varga’s existing audit
Edge deployment	Full acquisition-to-decision measurements, not parameter count alone

The strongest experiment is:

At matched event recall, does quality-aware monitoring reduce false welfare-check alerts in unseen environments?

Do not optimize false-alert reduction alone: a system that never alerts would win that metric.

5. Your next steps

Days 1–3: repair the evidence

Ask the agents to produce a claim-level evidence ledger, not another narrative synthesis.

Each important claim must contain:

- Claim ID and exact wording.
- Paper title, DOI, version, and corpus filename.
- Supporting quotation and page/section/table/equation.
- Hardware, antenna configuration, participants, rooms, and recording protocol.
- Ground-truth method and split procedure.
- What was measured versus inferred.
- Limitations and contradictory evidence.
- Status: supported, qualified, contradicted, or unverified.

Prioritize these claims:

1. The displacement ranges and claimed decibel deficit.
2. Absence of unconstrained heartbeat-identity demonstrations.
3. Exact WiDetect statistic and whether it measures motion or stationary occupancy.
4. Power Decline Ratio definition and evidence under non-fall confounders.
5. Sample-entropy liveness discrimination.
6. Raw complex CSI availability in each dataset.
7. Session/day/room metadata availability.
8. Exact open-set rejection method and prior art.
9. BabyMamba version and numerical discrepancies.
10. Exact CSI capability of your intended hardware.

A useful agent directive is:

Do not infer impossibility from absence of papers. Do not transfer results between tasks or hardware without justification. Separate source findings, engineering hypotheses, and unverified design choices. Locate counterevidence for every major claim.

Fix the attached specification's exclusion criteria too. Papers with weak evaluation belong in an audit corpus; exclude their results from strong deployment conclusions, not automatically from the review.^[1]

Days 4–7: pass the feasibility gate

Before freezing dataset partitions, inspect representative files.

Produce:

1. A hardware capability sheet: exact board/NIC, firmware, CSI dimensions, coherent RX-chain count, timestamps, packet-rate distribution.
2. A dataset capability sheet: representation, labels, participant IDs, room IDs, recording IDs, duration, reference signals, license.
3. A minimal capture script and raw-data diagnostic.
4. A documented decision between coherent-ratio acquisition and ESP32-native amplitude sensing.

Stop the ratio track if you cannot obtain suitable coherent measurements. Stop physiological-rate claims if you cannot obtain synchronized reference data.

Do not silently invent LOSO/LOEO groupings when provenance is missing.

Week 2: build falsification pilots

Start with a small pilot—not your final dataset. For example, use several consenting adults across two rooms, two days, and multiple placements to answer whether the pipeline is viable.

Test situations designed to break the proposed logic:

Situation	Desired interpretation
Empty room with fan or moving curtain	Do not mistake periodic interference for respiration
Quiet seated person	Do not transition to vacant merely because motion stops
Quiet person at a poor sensing position	Unobservable, not respiratory arrest
Quick sitting or lying down	Do not automatically label as fall
Suspicious descent followed by breathing	Do not automatically clear the event
Packet dropout or disconnected receiver	Sensor failure/unknown
Adjacent-room movement	Measure spatial false triggers
Two people	Reject unsupported single-person physiological interpretation

Do not ask elderly participants to fall or simulate medical distress. Any physical descent/fall study needs an approved, appropriately supervised protocol; begin with safe everyday transitions.

Weeks 3–4: establish baselines

Implement only:

- Statistical motion baseline.
- Spectral/autocorrelation respiration baseline.
- Small CNN or causal TCN.
- Fixed inactivity policy.
- Quality-aware policy.

Then compare:

Variant	Purpose
Amplitude + ordinary training	Basic reference
Alternative front end + same training	Isolate front-end benefit
Generalization method only	Isolate cross-domain learning
Quality gating only	Isolate selective sensing
Generalization + quality gating	Test combined contribution

Freeze partitions before windowing. Fit normalization, feature selection, calibration, rejection thresholds, and model selection using training/validation data only.

Audit preprocessing causality too. A filter that uses future samples may be acceptable offline but cannot support an immediate online decision without its buffering delay being disclosed.

Weeks 5–6: continuous evaluation

Public activity clips cannot substitute for continuous FSM validation. WiMANS, for example, consists of scripted three-second samples and excludes falls. ^[11]

Collect continuous normal-operation recordings with event timestamps and report:

- False welfare-check alerts per monitoring hour.
- Suspicious-event recall.
- Detection delay.
- Stationary occupancy recall.
- Fraction of time respiration is measurable.
- Respiratory-rate error only on valid-reference windows.
- Abstention rate and coverage–risk curves.
- Results by room, day, placement, orientation, and participant.
- Uncertainty intervals grouped by independent sessions/participants—not treating overlapping windows as independent evidence.

If your data contain healthy adults and simulated scenarios only, state that plainly. You cannot conclude that the system distinguishes benign rest from actual medical emergencies.

Weeks 7–8: profile and decide

Measure the complete pipeline:

$$T_{\text{decision}} = T_{\text{observation}} + T_{\text{preprocessing}} + T_{\text{inference}} + T_{\text{policy}}.$$

Also measure peak memory, input buffers, model storage, packet drops during inference, power, and sustained operation. Add ESP32 deployment only after confirming the acquisition/inference architecture is compatible.

Your next milestone should be a feasibility report containing a corrected evidence ledger, hardware proof, dataset capability matrix, pilot failure analysis, and initial baselines—not a polished chapter defending the current architecture.

For the next verification pass, provide the agents' evidence ledger and 93-paper manifest with filenames, titles, and DOIs. Those would let me check whether each claimed conclusion actually follows from your corpus rather than merely sounding plausible.

✱

1. [program.md](#)
2. https://www.eng.auburn.edu/~szm0001/papers/PhaseBeat_ACMHealth20.pdf
3. <https://github.com/xyanchen/WiFi-CSI-Sensing-Benchmark>
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